The RSOPT Project

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Chapter 6

Quadratic Programs

This chapter covers various forms of **quadratic programs** supported by the RSOPT project. This document defines quadratic programs as convex programs whose objective has a constant Hessian with no negative eigenvalues.

6.1 Unconstrained Quadratic Programs

Unconstrained quadratic programs (`UnconstrainedQuadraticProgram<N>`) are convex programs of the form

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top A x + b^\top x \tag{6.1}$$

where $A \in \mathbb{R}^{n \times n}$, $A \succeq 0$, and $b \in \mathbb{R}^n$. The relevant stationarity condition is

$$Ax^\star + b = 0 \Leftrightarrow x^\star = -A^{-1}b. \tag{6.2}$$

We solve unconstrained quadratic programs using the **conjugate gradient** (CG) method.

Definition. Two vectors p_i and p_j are *conjugate* with respect to A if $p_i^\top A p_j = 0$.

CG solves an n-dimensional system in at most n iterations by selecting conjugate search directions. The gradient of the objective at iterate x_k is

$$\nabla J_k = Ax_k + b. \tag{6.3}$$

CG selects $x_0 := 0$ and $p_0 := \nabla J_0 = b$ as the first search direction. Subsequent search directions p_k ($k > 0$) must be conjugate with all preceding search directions with respect to A. We write the recurrence as

$$p_{k+1} = Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i \tag{6.4}$$

and constrain β_i using the definition of conjugate vectors. For all $j = 0, 1, \cdots k$, we have

$$(Ax_{k+1} + b + \sum_{i=0}^{k} \beta_i p_i)^\top A p_j = 0. \tag{6.5}$$

Expanding gives us

$$x_{k+1}^\top A^\top A p_j + b^\top A p_j + \sum_{i=0}^{k} \beta_i p_i^\top A p_j = 0. \tag{6.6}$$

Notice that, by our "inductive hypothesis", $p_0, p_1, \cdots p_{k-1}$ are already pairwise conjugate with respect to A. We can now focus only on the case $j = k$ and take all other $\beta_i = 0$.

$$x_{k+1}^\top A^\top A p_k + b^\top A p_k + \beta_k p_k^\top A p_k = 0. \tag{6.7}$$

This gives us a closed form for β_k.

$$\beta_k = -\frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} \tag{6.8}$$

Therefore, the step direction p_k for $k > 0$ is given by

$$\boxed{p_{k+1} = \nabla J_{k+1} - \frac{(\nabla J_{k+1})^\top A p_k}{p_k^\top A p_k} p_k.} \tag{6.9}$$

After selecting the step direction, it is necessary to select the step size α_k. We select α_k according to the QP

$$\min_{\alpha_k \in \mathbb{R}} \frac{1}{2}(x_k + \alpha_k p_k)^\top A (x_k + \alpha_k p_k) + b^\top (x_k + \alpha_k p_k) \tag{6.10}$$

Expanding the objective and dropping all terms not dependent on α, we have

$$\frac{d}{d\alpha_k}\left[\alpha_k p_k^\top A x_k + \frac{1}{2}\alpha_k^2 p_k^\top A p_k + \alpha_k b^\top p_k\right] = 0. \tag{6.11}$$

We find that α_k is also determined analytically as

$$\boxed{\alpha_k = \frac{-(\nabla J_k)^\top p_k}{p_k^\top A p_k}.} \tag{6.12}$$

The complete CG algorithm is detailed in Algorithm 1. For two conjugate directions p_i and p_j with respect to A, we can interpret this algorithm in terms of the eigendecomposition of A.

$$p_i^\top A p_j = p_i^\top S \Lambda S^\top p_j \tag{6.13}$$

We can interpret the algorithm as akin to taking orthogonal steps to minimize an objective of the form $x^\top \Lambda x$, where Λ is a diagonal matrix containing the eigenvalues of A.

Algorithm 1 Conjugate Gradient

Require: A, b

$x_0 \leftarrow 0$
$p_0 \leftarrow b$
$g_0 \leftarrow b$
$\alpha_0 \leftarrow -g_0^\top p_0 / p_0^\top A p_0$
$x_1 \leftarrow x_0 + \alpha_0 p_0$
for $k = 1 \cdots n-1$ **do**
    $g_k \leftarrow A x_k + b$
    $p_k \leftarrow g_k - (g_k^\top A p_{k-1} / p_{k-1}^\top A p_{k-1}) p_{k-1}$
    $\alpha_k \leftarrow -g_k^\top p_k / p_k^\top A p_k$
    $x_{k+1} \leftarrow x_k + \alpha_k p_k$
end for
return x_n

6.2 Equality-Constrained Quadratic Programs

TODO

6.3 Inequality-Constrained Quadratic Programs

TODO